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1 Classic binomial and multinomial identities

Definition 1.1. Factorial of n , denoted by $n!$, is the number of ways to put n distinct objects in a queue.

Definition 1.2. Binomial coefficient $\binom{n}{k}$ is the number of ways to select k distinct objects from n distinct objects.

Definition 1.3. For $n = a + b + c$ the multinomial coefficient $\binom{n}{a,b,c}$ is the number of ways to split n distinct objects into different boxes, where the first box contains a objects, the second box contains b objects, the third box contains c objects.

One may create multinomial coefficient for any number of boxes :)

Exercise 1.1. Is $\binom{n}{k}$ equal to $\binom{n}{n-k}$?

Is $\binom{n}{k}$ equal to $\binom{n}{k,n-k}$?

Exercise 1.2. Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

Exercise 1.3. Write

$$\binom{n}{a,b,c,d}$$

using binomial coefficients.

Write

$$\binom{n}{a,b,c,d}$$

using factorials of n , a , b , c and d .

Idea 1. Solution strategies for combinatorial identities:

1. Double counting or story proof.
 2. Visual identity (rare gem!).
 3. Evaluation of a polynomial at a special point. Differentiation, integration.
 4. Combine old identities into new.
 5. Guess the answer and verify by induction.
 6. Dark art!
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Pascal's triangle and rule

Exercise 1.4. Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

Exercise 1.5. Pascal's rule. Simplify

$$\binom{n}{k} + \binom{n}{k+1}.$$
$$\binom{n}{a-1, b, c} + \binom{n}{a, b-1, c} + \binom{n}{a, b, c-1}.$$
$$\binom{n}{a-1, b, c, d} + \binom{n}{a, b-1, c, d} + \binom{n}{a, b, c-1, d} + \binom{n}{a, b, c, d-1}.$$

Exercise 1.6. Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

Exercise 1.7. Simplify

$$\sum_{a+b+c=2026} \binom{2026}{a, b, c}.$$
$$\sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}.$$
$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$
$$\sum_{a+b \leq n} \binom{n}{a} \cdot \binom{n-a}{b}.$$

Here a , b and c are assumed non-negative integers.

Hockey-stick identity

Exercise 1.8. Hockey-stick identity, Christmas stocking identity, boomerang identity.

Simplify

$$\sum_{n=a}^b \binom{n}{a}.$$
$$1 + 2 + 3 + \cdots + n.$$
$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1).$$
$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2).$$

Exercise 1.9. Obtain closed form expressions for:

$$1^2 + 2^2 + \cdots + n^2,$$

$$1^3 + 2^3 + \cdots + n^3,$$

$$1^4 + 2^4 + \cdots + n^4.$$

Exercise 1.10. Hockey-sticks plus idea from linear algebra!

Decompose $n^2 + 2n + 5$ using $\binom{n}{0}$, $\binom{n}{1}$, $\binom{n}{2}$ as the basis.

Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

Using similar trick find the sums

$$\sum_{k=1}^n 5k^2 + 4k, \quad \sum_{k=1}^n k^3 + 2k^2.$$

Exercise 1.11. Let's start from some number $\binom{n}{k}$ inside the Pascal's triangle.

In one step we can go either in north-east or north-west direction.

Find the sum of all numbers we can reach.

Wandermonde's identity

Exercise 1.12. Simplify

$$\binom{a}{0} \binom{b}{m} + \binom{a}{1} \binom{b}{m-1} + \binom{a}{2} \binom{b}{m-2} + \cdots + \binom{a}{m} \binom{b}{0}.$$

$$\binom{a}{0} \binom{m}{0} + \binom{a}{1} \binom{m}{1} + \binom{a}{2} \binom{m}{2} + \cdots + \binom{a}{m} \binom{m}{m}.$$

$$\binom{a}{0}^2 + \binom{a}{1}^2 + \binom{a}{2}^2 + \cdots + \binom{a}{a}^2.$$

Exercise 1.13. There are $n = 500$ fishes in the pond, $t = 30$ of them are tagged. You capture $c = 5$ fishes at random without replacement. Let the random variable Y be the number of captured tagged fishes. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 1.14. There are $n = 24$ candies in the jar, 14 with orange flavour and 10 with ananas flavour. You take 6 candies at random. Let Y be the number of candies with orange flavour you picked. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 1.15. A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

What is the probability that the hand contains no spades?

What is the probability that the hand contains exactly 5 hearts?

What is the probability that the hand contains at most two aces?

Exercise 1.16. Simplify

$$\sum_{a+b+c=k} \binom{n_1}{a} \binom{n_2}{b} \binom{n_3}{c}.$$

Binomial or Taylor or MacLaurin expansion

Exercise 1.17. Write the Taylor expansion (binomial expansion) at $x = 0$ for

$$(1+x)^n, \quad (x+y)^n.$$

Exercise 1.18. By plugging some special points to the $(x+y)^n$ or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$
$$\sum_{k=0}^n k \binom{n}{k}, \quad \sum_{k=0}^n (k-1) \binom{n}{k}, \quad \sum_{k=0}^n (-1)^k \binom{n}{k}, \quad \sum_{k=0}^n (-1)^{k-1} k \binom{n}{k}$$

Hint: differentiation may help you.

How that can be?

Exercise 1.19. Be brave, really brave!

Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/2}{k}, \quad \binom{-1/2}{n}.$$

Exercise 1.20. Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

Draw the Pascal's triangle starting from $\binom{-1/2}{0}$.

Draw the Pascal's triangle starting from $\binom{1/3}{0}$.

Exercise 1.21. Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

Exercise 1.22. Simplify

$$\binom{10}{0} \binom{-0.5}{7} + \binom{10}{1} \binom{-0.5}{6} + \binom{10}{2} \binom{-0.5}{5} + \cdots + \binom{10}{7} \binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of $(1+x)^1, (1+x)^{-0.5}, (1+x)^{10-0.5}$.

Exercise 1.23. Homage to Newton.

Write the Taylor expansion at $x = 0$ for $(1-x^2)^{1/2}$.

Visually (or analytically if you wish) find the integral

$$\int_{-1}^1 (1-x^2)^{1/2}.$$

Calculate the sums

$$\binom{1/2}{0} - \frac{1}{3} \binom{1/2}{1} + \frac{1}{5} \binom{1/2}{2} + \cdots$$
$$\sum_{k=2}^{\infty} \frac{1}{2k+1} \frac{(2k-3)!!}{(2k)!!}$$

Gauss summation trick

Exercise 1.24. Simplify

$$1 + 2 + 3 + \cdots + n;$$

$$1^2 + 2^2 + 3^2 + \cdots + n^2;$$

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n};$$

$$\binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \cdots + (n+1)\binom{n}{n};$$

$$\binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} + \cdots + (n-1)\binom{n}{n};$$

$$\binom{n}{0} + 3\binom{n}{1} + 5\binom{n}{2} + \cdots + (2n+1)\binom{n}{n};$$

$$\binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} + \cdots + (-1)^n(n+1)\binom{n}{n};$$

$$3\binom{n}{1} + 7\binom{n}{2} + 11\binom{n}{3} + \cdots + (4n-1)\binom{n}{n};$$

$$\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - 4\binom{n}{4} + \cdots + (-1)^n n\binom{n}{n};$$

First day quiz :)

1. How many ways are to order 6 persons in a queue?
2. The box contains 50 balls numbered from 1 to 50. We draw 6 balls without replacement and write down the resulting sequence of numbers. How many different sequences are possible?
3. I have 5 yellow books and 7 red books. I want to place them on my bookshelf and I would like to place yellow books to the left of red books. How many book orders are possible?

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- [Gra94] Ronald L Graham. Concrete mathematics: a foundation for computer science. Pearson Education India, 1994. A classic captivating (really funny sometimes!) treatise on combinatorics from masters, 690 pages.
- [Har+10] John J Harris et al. Combinatorics and graph theory. Springer-Verlag New York, 2010. Accessible book with a lot of pictures, stories and nice explanations, 355 pages.
- [Gui25] David Guichard. An Introduction to Combinatorics and Graph Theory. 2025. URL: https://www.whitman.edu/mathematics/cgt_online/cgt.pdf. Short open access introductory book with pictures and good intuition, 155 pages.
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